Work Package: [CAN-FD Sniffer (CANSNIF)]

Version: 1

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1. Scope & Objective

- **Purpose:** An STM32-based sensor board that logs communication over CAN-FD into an micro SD CARD. Battery backed RTC 2032 or other common batteries. 2032 can be mounted on the backside as it is a big big.
- **Key Features:** CAN-FD / micro SD-CARD

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2. Mechanical Specifications

Feature	Specification	Notes
Board Dimensions	50mm x 50mm	Standard 50x50 agreed layout
Mounting Holes	4 × M3	Standard 50x50 agreed layout
Hole Offsets	5mm from edges, dx = 5, dx = 5	Standard 50x50 agreed layout
Height	Potential battery slot on the back side.	Double check if populated.
Antennas	No	Not present, so no constraints about the enclosure
Pluggable Inserts	SD CARD	Any enclosure must have a way to get the SD card in and out

See [board dimensions].

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3. Electrical Architecture

3.1 Microcontroller (MCU)

- **Model:** STM32U575RGTx !LDO Version
- **Max Clock Speed:** 160MHz
- **Operating Voltage:** 3.3V
- **Available devkit in lab:** no
- **Read The Functioning Manual**: has internal LDO with external capacitors.
- 4-48 MHz crystal. Check with Cosmin exactly what 16Mhz SMD we have in the lab
- 32k crystal needed for RTC

3.2 Board setup

4 layer board with SIG,GND,VCC,SIG

3.3 Communication Interfaces

- **CAN-FD**
- **SWD**
- **SDMMC - SD Card**
- **(Optional) - USB Type C.**

  Protocol Table:

Interface	Speed/Baud	Pins	Notes
CAN-FD	Max - must be able to efficiently log for worst case scenario.	CRX CTX	Lots of other options, makes for easier layout. TJA 1051T/3 transceiver. Must have a 120 ohm termination resistor that can be enabled using a solder bridge or jumper (see nucleo schematic) CTRL+LCLICK in CUBEMX to see other available pins. Silent mode.
SDMMC	?	?	4 bit wide bus
USB	2.0 FS	?	Optional, if it fits on the board. Watch out for unintended power flow.

			E.G. Laptop -> This Board -> All of the 5V rail. Board VCC should be supplied though a diode from 5V rail and one from USB.

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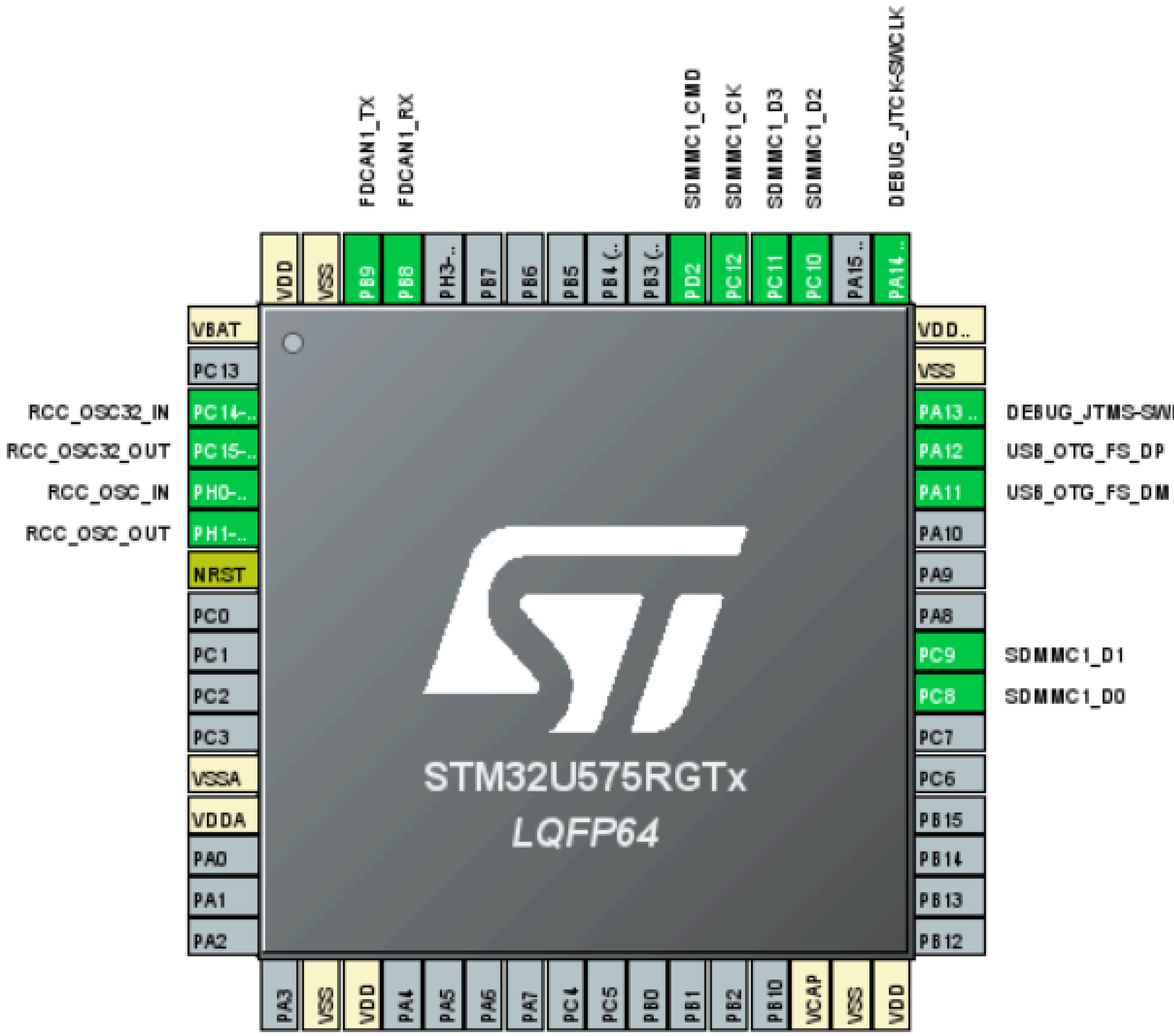
4. GPIO & Signals Mapping

- **4.1 Digital Outputs:**
  - One error LED pin for full sd card or error state.
  - One LED for SD writing activity.



Suggestion: CubeMx package also in folder

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5. Power Requirements

- **Input Voltage (Vin):** 5V
- **Regulated Rails:** 5V, 3.3V
- **Estimated Peak Current:** 200mA-300mA, any LDO will do just fine. Add a decent bulk capacitor for SD.

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6. Connector Definitions

- **Power In:** JST XH - 2p. Pin1: 5V
- **Debug:** 5 x 2.54 male header: Pin1: 3v3, SWDIO,SWDCLK, NRST, GND
- **CAN-FD:** JST-XH: PIN1: CANH, PIN2, CANL, PIN3: GND
- **SD-Conn**: Something robust against vibrations and shocks.
- **(Optional) USB:** GF4105 type C.
- Silkscreen indication where it's easy to connect backwards. If it can fail like that, it will fail like that.

Note: Suggested connectors, probably not ideal.Double check if present on other boards

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7. Deliverables & Milestones

1. **Schematic Review:** Week 22-28 February
2. **Layout & DRC Check:** Soon
3. **Gerber Submission:** Soon
4. **BOM:** Soon
5. **Assembly & Smoke Test:** Soon
6. **Nice pinout image:**

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8. Software Package

Github link when board is done and functional.

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9. Useful links

1. SD Card for STM32
   https://deepbluembedded.com/stm32-sdmmc-tutorial-examples-dma/
2. Product page
   https://www.st.com/en/microcontrollers-microprocessors/stm32u575cg.html

Notes:

1. If stuck, ask around.
2. Check for common components that we have. No need for 5 types of CAN transceivers.
3. Use KiCad 9, preferably the latest version.

4. This design and documentation will be open sourced
5. KISS - keep it simple stupid
6. This is a project to gain experience and knowledge. It will be a very good board or a very good smoke machine, have fun.